Semantic-DP Distillation: Choosing the Granularity of Client-Private Supervision

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ABSTRACT

Strong public features leave a private design question: at what granularity should clients release task supervision? Per-example votes spend sensitivity on every public query, while a regional target reuses supervision across related examples. Semantic-DP Distillation (SDD) releases one bounded class vote per public semantic region, yielding KC coordinates and tight replace-one whole-client sensitivity $\sqrt{2K}$. We characterize regional target quality through an exact purity–decoding decomposition and a margin-dependent noise bound, and use a public-only approximation to choose release resolution. At taxonomy resolution, weighted matching provides an alternative decoder that preserves class coverage without an additional privacy release. On frozen public representations with 10^4 strongly heterogeneous CIFAR-10 clients, matching reaches 86.3% accuracy at $\varepsilon = 0.1$, $\delta = 10^{-5}$ without public task labels, confirmed by an independent five-seed group, using 200× fewer report coordinates than per-example transfer. Natural-writer and natural-author studies characterize how public geometry and participation govern the choice of release. Compact head averaging supplies higher accuracy when its parameter release is supported by the cohort and budget. SDD makes semantic reuse a measurable resource for private supervision.

CCS CONCEPTS

• **Security and privacy** → **Privacy-preserving protocols**; • **Computing methodologies** → *Machine learning*.

KEYWORDS

differential privacy, federated learning, distributed DP, knowledge distillation, semantic partition, private aggregation

1 INTRODUCTION

Public representations can encode useful geometry before any private task labels are available. In community-content learning, users retain labelled post histories and local models; a coordinator supplies a public feature map, an unlabelled proxy and a class taxonomy. Protected aggregation can collect bounded contributions without exposing individual reports [5, 6, 10]. The learning question is what supervision those contributions should carry. With strong public features, the choice is between releasing parameters, querying every proxy example, and reusing one target over a semantic region.

The gap: release resolution needs a utility criterion. PATE aggregates teacher votes [25, 26], and federated distillation transfers predictions on public examples [2, 16, 20, 22]. A joint per-example vote report has replace-one whole-client sensitivity $\sqrt{2|P|}$, even when related queries share a useful target. A coarser release reduces sensitivity but can assign a common target to examples from different classes. A release rule must account for both public geometry and the noise that the actual cohort can average. Public representations are an explicit resource: prior work establishes their importance for private learning [21, 29]. Our question concerns the private supervision exchanged on top of them.

The idea: choose the unit of private supervision. Semantic-DP Distillation (SDD) forms K public regions and releases one class vote per client per region. Every proxy example inherits its region's target. The report has KC coordinates and tight whole-client sensitivity $\sqrt{2K}$, independently of proxy size at fixed K. A conditional quality model separates regional impurity from target decoding and explains when additional budget stabilizes targets, when additional regions supply useful resolution, and when the cohort cannot support that resolution. A public-only predictor evaluates candidate K without consulting private margins or final-test labels. At $K = C$, weighted matching offers a coverage-constrained decoder using the same protected matrix.

The evidence: compact supervision with a verified mechanism. At actual 10k strongly heterogeneous CIFAR-10 clients, $\varepsilon = 0.1$ and $\delta = 10^{-5}$, a fixed $K = C = 10$ matching decoder reaches 86.3% **accuracy without public task labels**, with five original and five independent confirmation seeds. Its 100-coordinate report uses 200× **fewer coordinates** than per-example DP-FedCT. On the original paired seeds, the gain over per-example transfer is 16.6 points (95% interval [14.6, 18.5]); independent local learning gives 42.4%. Replaying regional targets identifies the mechanism: matching repairs repeated-class assignments in three of ten seeds and leaves the other seven endpoints unchanged.

Matched partitions and query controls separate semantic reuse from simply reducing the number of queries. A Gaussian approximation predicts regional release-label quality across participation, resolution and privacy budgets, validated against independent exact distributed noise. Distinct Usenet authors and official Federated EMNIST writers supply natural client boundaries. Together with a compact DP-Head baseline, these studies characterize which released object the available geometry and aggregation resources support.

Contributions.

- **A regional interface for whole-client private supervision.** SDD reuses targets across public examples with a KC-coordinate report and $\sqrt{2K}$ sensitivity. Established distributed integer noise protects its adaptive transcript under an aggregate-only interface (§3).
- **A quality criterion for release granularity.** An exact regional-label decomposition, an uncentered margin bound and an independently checked Gaussian approximation connect public geometry to noise-supported resolution (§4).
- **Conditional decoding and empirical release choices.** Paired decoder controls, independent seed confirmation, and

natural client studies identify when semantic reuse, class coverage and parameter averaging supply useful private supervision (§5).

2 RELATED WORK

Private parameters and public representations. DP-SGD clips and noises example gradients [1]; client-private federated averaging applies bounded contributions to whole clients [18, 23]. Secure aggregation and distributed discrete Gaussian noise remove a trusted curator from the release interface [6, 9, 17]. DP-FTRL uses correlated noise and a different training/accounting construction [19]; our matched parameter baseline uses the same independent-round distributed mechanism as the vote interfaces. Tramèr and Boneh [29] show the value of public features for private learning, and Li et al. [21] establish the strength of private adaptation of pretrained language models. We hold the learner's public representation fixed when comparing release objects and change partition features separately in structure controls.

Federated distillation and private queries. FedMD shares public-query predictions across heterogeneous models [20]; FedDF distils an ensemble into a server model [22]; DS-FL exchanges and sharpens public-example outputs [16]. These transfer mechanisms motivate prediction exchange, while our comparisons calibrate every released object to replace-one whole-client adjacency. PATE protects noisy teacher-vote answers [25, 26], and Private-kNN uses local neighbourhoods for private query labelling [32]. FedCT exchanges hard labels with local randomisation [2]. Our DP-FedCT control keeps its local co-training schedule and replaces the release layer with the same protected joint vote sum as SDD. Its singleton regions isolate release resolution. PATE-DI additionally receives protected confidence screening, charged to the same whole-run budget. Recent personalized distillation at this venue addresses knowledge calibration across clients [30].

Cluster reuse and the protected unit. NoiseCluster [27] reuses a noisy cluster majority under label privacy. Its adjacency changes one label with fixed features, giving $\sqrt{2}$ sensitivity for the cluster-count vector. Replacing one complete SDD client can change all K model responses, giving $\sqrt{2K}$. SDD studies regional model reports in adaptive client collaboration, with a quality criterion and distributed whole-run privacy. The numerical privacy budgets of these different protected units are not directly interchangeable.

Protected Web aggregates. Prio, Prochlo and split-and-mix protocols support aggregate-only interfaces [5, 10, 15]. Browser reporting and summary releases motivate bounded contributions [3, 12, 14]; distributed noise sharing also appears in Web data markets [31]. Shuffling offers an alternative amplification route [11, 13]. SDD's accountant operates on the protected noisy sum. Its report-coordinate counts quantify the learning payload before protocol overhead.

3 SEMANTIC-DP DISTILLATION

A public semantic region lets one released target supervise many proxy examples. SDD uses regions as the unit of private knowledge exchange: clients cast one bounded vote per region, and the protected aggregate supplies shared targets for local learning.

Table 1: Release interfaces under replace-one client adjacency, before quantisation and transport sharing. B is the clipping norm.

released object	coordinates	L_2 sensitivity
clipped parameter update	d_{model}	$2B$
joint per-example votes	$\lvert P\rvert C$	$\sqrt{2\lvert P\rvert}$
public-region votes	KC	$\sqrt{2K}$

Table 1 compares clipping and release resolution. Our compact head baseline shares 128-dimensional features. SDD requires compatible class opinions while keeping each client's parameters local.

3.1 A public partition for knowledge reuse

A fixed public extractor g and public K-means form nonempty regions $P_1, \ldots, P_K$, shared across rounds independently of private data. At fixed K, additional proxy examples increase distillation inputs without increasing reports. Singleton regions at $K = |P|$ recover per-example transfer. Semantic reuse requires compatible task information within a region. Random partitions satisfy the same sensitivity bound; matched experiments test the learning value of their geometry separately.

3.2 One bounded opinion per region

For client model f_i, the regional posterior is

$$\bar{R}_i[k, \cdot] = \frac{1}{|P_k|} \sum_{x \in P_k} f_i(x) \in \Delta^{C-1}. \tag{1}$$

SDD-M votes for the largest component of $\bar{R}_i[k, \cdot]$. SDD-R votes on $f_i(x_k^*)$, where x_k^* is the public point nearest the centroid: public representative queries followed by regional label propagation. Both emit K one-hot rows and propagate targets over identical regions. Fixed-K comparisons distinguish pooled and localized responses while holding sensitivity, report size and partition quality fixed.

Reports lie in a product of K probability simplexes. Direct posteriors also satisfy this bound, with quantisation slack. The evaluated one-hot encoding lies exactly on the integer grid. Common learners isolate release granularity; the interface itself requires compatible class outputs from local models.

3.3 Noisy consensus and local learning

Each client adds independent integer discrete-Gaussian shares to its quantised report. The aggregate-only layer exposes $S = \sum_i (sR_i + G_i)$, and the server projects $S/(sn)$ row-wise onto the simplex to obtain A. For ordinary consensus, the target is

$$\hat{y}(x) = \arg\max_c A[k(x), c]. \tag{2}$$

We also evaluate a fixed class-prior correction: with public region weights $w_k = |P_k|$, subtract $\pi_c = \sum_k w_k A[k, c] / \sum_k w_k$ before taking the arg-max. This uses the released aggregate and public sizes. It can counter a common class preference when clients have narrow local label coverage. Public-configured runs select ordinary or centered consensus with public development data, shared across label-release methods. The fixed condition uses centered consensus without public task labels.

Coverage-constrained decoding. At taxonomy resolution $K = C$, a prespecified alternative is a one-to-one assignment using the released matrix:

$$\hat{\pi} = \arg\max_{\pi \in \mathfrak{S}_C} \sum_k |P_k| A[k, \pi(k)], \qquad \hat{y}(x) = \hat{\pi}(k(x)). \tag{3}$$

Hungarian assignment solves this public-size-weighted objective. It guarantees class coverage among regional targets, while correctness depends on whether the public regions support a one-to-one class allocation. Equal numbers of regions and classes alone do not establish that condition. The decoder requires no further private query or privacy budget. We also test $A[k, \cdot]$ as a soft training target and a fixed-temperature softmax of centered scores. These are proportions of hard client votes, rather than calibrated class posteriors. All variants use the same protected release, with their full paired results retained.

Later-round batches mix equal private and proxy halves, with hard or soft cross-entropy according to the declared decoder and fixed optimisation steps. Every proxy item inherits its region's target. Three releases are charged; releases 1/2 supervise the next training rounds, and release 3 follows the final local fit. The confidence-filtered baseline samples its answered subset under separately budgeted screening.

Report size. At $|P| = 2000$, $C = 10$, a $K = 10/20$ release uses 100/200 integer coordinates, versus 20000 per-example coordinates and 1280 head coordinates. Counts measure per-round learning payload before secret sharing and cryptographic overhead.

Choosing resolution without private tuning. The public design criterion in §4 scores release-label quality using public teachers and deployment noise. Original public-configured grids retain their frozen pilot settings; the new criterion is evaluated separately. Fixed checks use $K = C$ and the declared decoder without public task labels. Both regional estimators remain separate. Final runs aggregate actual client reports and independent shares, calibrated for that same population.

3.4 Protected aggregate and whole-run client privacy

A fixed cohort of n clients participates in each of T rounds. Replace-one adjacency changes the entire dataset of one honest client. Public features, partitions and configurations are independent of private datasets. The adversary sees the noisy sum and broadcasts; local models, individual reports and honest noise shares remain hidden by the aggregation interface.

Aggregation assumptions. A1: secure additive aggregation reveals the sum and metadata that is public or independent of protected data. A2: at most $t < n$ contributors disclose reports and noise to the server, leaving $h = n - t$ honest independent shares. A3: aggregation parties preserve that view against the stated coalition. A4: honest contributors use bounded reports and fresh independent noise; participation is fixed ($q = 1$), with recalibration or abort if the required honest cohort is unavailable. Secret-sharing or split-and-mix protocols can instantiate the interface [6, 10, 15]. Experiments implement protected-sum arithmetic; transport is a deployment component. The guarantee concerns observation and collusion under A1–A4.

Lemma 3.1 (Report sensitivity). *For $R, R' \in (\Delta^{C-1})^K$, $\|R - R'\|_2 \le \sqrt{2K}$, with equality for opposing one-hot rows. This holds for every public partition.*

Lemma 3.2 (Integer sensitivity). *One-hot reports have $\Delta_q = s\sqrt{2K}$ exactly. Rounded posteriors have $\Delta_q \le s\sqrt{2K} + \sqrt{KC}$. A parameter update clipped at B and guarded after rounding in the integer ball $L_B = \lceil sB + \sqrt{d}/2 \rceil$ has $\Delta_q \le 2L_B$.*

Release. Each client submits shares of $sR_i + G_i$ modulo 2^{64}, with $s = 2^{16}$ and independent coordinates $G_i[j] \sim \mathcal{N}_{\mathbb{Z}}(0, b^2)$. The server projects the dequantised sum onto row simplexes and applies the specified decoder. The exact discrete-Gaussian sampler and integer guards are specified in Appendix C.

Theorem 3.3 (One protected aggregate). *Under A1–A4, the aggregate in dimension d is ρ_r-zCDP, where*

$$\rho_r = \frac{\Delta_q^2}{2hb^2} + \frac{d\tau_h(b)}{4}, \quad \tau_h(b) \le 10(h-1)e^{-\pi^2 b^2}. \tag{4}$$

For $h = 1$, $\tau_h = 0$; Appendix B gives its exact expression.

This applies the convolution guarantee of Kairouz et al. [17] to the regional integer query after the coalition subtracts its disclosed shares. Additional honest noise, modular addition, projection, matching and soft-target construction preserve privacy.

Theorem 3.4 (Adaptive whole run). *Conditionally calibrated releases compose to $\rho_{\text{total}} \le \sum_{r=1}^{T} \rho_r$. The complete server transcript is (ε, δ)-DP whenever*

$$\rho_{\text{total}} + 2\sqrt{\rho_{\text{total}} \log(1/\delta)} \le \varepsilon. \tag{5}$$

Condition on data-independent training random tapes and each preceding public transcript: another client's unchanged private data and that transcript fix its response; the replaced client's report stays within the same bounded space. Adaptive composition therefore includes every broadcast [7]. Public-only design consumes no client budget. Private local models are not released. Test metrics and clean-count diagnostics are offline measurements, outside the deployment transcript claimed to satisfy this guarantee.

Writing $L = \log(1/\delta)$ and $\rho_\varepsilon = (\sqrt{L+\varepsilon} - \sqrt{L})^2$, we round b upward from $\Delta_q / \sqrt{2h\rho_\varepsilon / T}$ and verify Eq. (4) including its convolution correction. Every release retains a numerical certificate with actual n, honest h, integer sensitivity, noise scale and composed budget. The averaging denominator and calibration always use the same participating population. For $t = 0$, mean-noise standard deviation is bounded by

$$v = \frac{b\sqrt{n}}{sn} \approx \frac{\sqrt{TK/\rho_\varepsilon}}{n}. \tag{6}$$

This scale connects release resolution to target quality in §4.

4 RELEASE RESOLUTION AND TARGET QUALITY

Choosing K balances the task resolution supplied by public geometry against the reliability of a noisy regional decision. We analyze the released target itself, the common supervision subsequently combined with private examples.

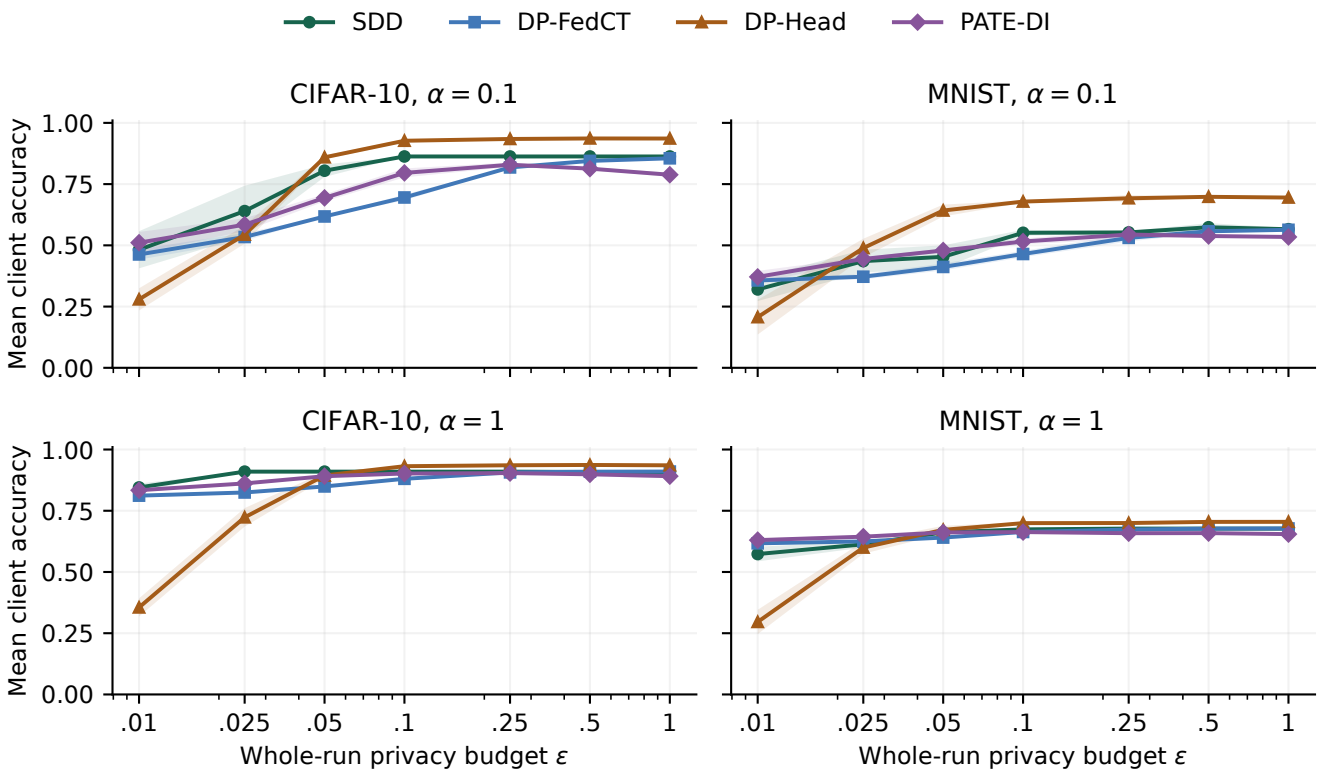


Figure 1: Original public-configured privacy–utility grids, actual 10k clients, three seeds with sample SD, seven budgets and two heterogeneity levels. SDD denotes mean regional voting with the original consensus decoder; matching is evaluated separately in Table 2. Methods share learner features and private optimisation. DP-Head releases a global head; PATE-DI includes protected screening costs. All configurations retain the original public-only selection.

4.1 An exact geometry–decoding decomposition

Fix a partition and a labelled diagnostic proxy. Let $w_k = |P_k|/|P|$, $a_{kc} = \Pr_P(y = c \mid k)$, $a_k^* = \max_c a_{kc}$, and $Q_{kc} = \Pr(\hat{y}_k = c \mid q)$ for clean mean regional votes q and the specified release/decoder. For any hard decoder, including coupled matching,

$$\mathbb{E}[\mathrm{Acc}_P] = \sum_k w_k \sum_c a_{kc} Q_{kc}. \tag{7}$$

Writing $\mathrm{Pur}(K) = \sum_k w_k a_k^*$ gives the exact identity

$$1 - \mathbb{E}[\mathrm{Acc}_P] = 1 - \mathrm{Pur}(K) + \sum_{k,c} w_k (a_k^* - a_{kc}) Q_{kc}. \tag{8}$$

The first term is the impurity of constant regional labels; the second measures the decoder's allocation among classes. It includes both clean teacher disagreement with the regional majority and noise-induced changes. Thus a flip probability alone is insufficient: an alternative class can still be correct for part of a mixed region.

Purity bounds a constant hard-label classifier on this same proxy. Final local models also receive private examples and make nonconstant predictions, so Eq. (7) is a release-label model, with local-model accuracy evaluated separately. Partitions fitted at different K need not be nested; both purity and teacher margins are measured for each candidate rather than assumed monotone.

4.2 Noise-supported regional margins

For one release, the mean-noise coordinate is sub-Gaussian with variance proxy $v^2 = nb^2/(s^2n^2)$, retaining exact integer calibration. Let $c_k = \arg\max_c q_{kc}$ and $m_{kc} = q_{k,c_k} - q_{kc}$.

Theorem 4.1 (Uncentered regional target bound). *For coordinate arg-max of the uncentered noisy vote matrix and $c \neq c_k$,*

$$\Pr(\hat{y}_k = c \mid q) \leq \exp[-m_{kc}^2/(4v^2)]. \tag{9}$$

Consequently the conditional expected proxy-label accuracy is at least

$$\sum_k w_k \left[a_{k,c_k} - \sum_{c \neq c_k} (a_{k,c_k} - a_{kc})_+ e^{-m_{kc}^2/(4v^2)} \right]_+ . \tag{10}$$

Independent discrete-Gaussian coordinates have sub-Gaussian parameter v; a competitor must overcome margin m_{kc} with a difference of parameter $\sqrt{2}v$. Chernoff's bound gives Eq. (9); summing only positive correctness losses gives Eq. (10). The theorem concerns unwrapped coordinate arg-max; simplex projection preserves its winning coordinate. The modular wrap probability

certified in Appendix B supplies an additive error bound for ring arithmetic. Centered decoding couples regions through the released prior; matching couples them through assignment, so these decoders use the general identity rather than the uncentered tail bound.

Without collusion, $v \approx \sqrt{TK/\rho_\varepsilon}/n$. For two classes under continuous independent Gaussian noise, the clean winner flips with probability $\Phi[-m_k/(\sqrt{2}v)]$. For multiple classes, we simulate all competitors and the actual projection and prior correction. This Gaussian approximation predicts Q_{kc} in Eq. (7); independent exact-noise trials check its accuracy. When all relevant margins dominate v, decoding stabilizes. Further budget then preserves regional targets; additional task resolution requires changing the partition or target representation.

4.3 Public-only resolution selection

The deployment rule trains teachers on labelled public development examples, forms their regional votes, and estimates a_{kc} on public validation examples assigned to the nearest public centroid. For each candidate K, it simulates Gaussian noise at the actual deployment certificate's v, applies centered decoding, and computes expected validation label quality using validation region frequencies as weights. It selects the largest expectation, breaking ties toward smaller K. This directly evaluates the geometry–noise tradeoff without private-count access. It is a public design surrogate, rather than an optimizer of final local-model accuracy. The no-public-task-label condition instead fixes $K = C$ and its decoder; benchmark labels enter offline diagnostics alone.

5 EVALUATION

The evaluation asks which released object supplies useful supervision under fixed public features, whole-client privacy and local computation. Target quality tests the resolution model; paired decoders test class allocation; natural clients and parameter baselines locate the supported operating points.

5.1 Experimental setup

Common representations and local resources. Image clients use frozen public DINOv2 ViT-S/14 features [24], PCA to 128 dimensions fitted on the public proxy alone, and row normalisation. Each independently initialised softmax head receives three rounds of 60 SGD steps, batch 128, learning rate 0.1, momentum 0.9 and weight decay 5×10^{-4}. Round 1 uses private data; later batches mix 64 private and 64 proxy items. Local-only uses private data for all 180 steps. Every method uses the same client data, initial state and private sampling streams within a seed. DP-Head also evaluates its local endpoint immediately before the third broadcast, with the same 180 steps. All three releases remain charged; no extra adaptation is supplied to either endpoint.

CIFAR-10/MNIST clients draw 600 examples with replacement using Dirichlet mixtures $\alpha \in \{0.1, 1\}$. The eligible private pools have 45000/53000 source images, after excluding proxy and development examples. Different constructed clients can reuse source images; adjacency protects one whole constructed dataset. The unlabelled proxy has 2000 examples and public development has 5000 labels. Proxy-excluded validation/test contain 3967/4033 CIFAR-10 and 5000/5000 MNIST examples. Constructed image clients study participation at controlled heterogeneity; the two natural datasets below retain original client boundaries.

Natural clients and disjoint evaluation. Usenet clients comprise all training posts associated with one From-header author identifier. We use six content families, remove headers and quoted lines from inputs, fit TF-IDF on public text, and fit 128-dimensional SVD on the proxy. There are 4249 eligible private authors, median one post each; 1500 proxy and 1500 development posts come from separate public authors. Validation/test have 1747/1737 posts from disjoint held-out authors. The predeclared Web metric is mean client balanced accuracy; standard accuracy and per-family recalls are retained.

Official Federated EMNIST [8, 28] supplies 3400 writers and 62 classes. A frozen seed-1729 split assigns 200 writers each to proxy, development, validation and test roles; 2600 remain private. Each selected private client keeps all of its original training records: 510725 records are available, with no source-record overlap between clients. Proxy/development contain 2000/5000 examples; validation/test contain 4685/4549 official test records from their own held-out writers. Thus writer roles are disjoint in addition to the dataset's original within-writer split. Images keep their official polarity and use the same frozen extractor and public-only PCA as the other image tasks. We evaluate 1000/2000 distinct writers, $\varepsilon = 0.1, 1$, reporting accuracy and macro recall.

Two configuration resources. The *fixed* condition uses $K = C$ and a declared decoder, with no public task labels for configuration or targets. The *public-configured* condition permits labelled development teachers and validation selection. Original curves retain the frozen pilot over $K \in \{5, 10, 20, 50, 100\}$, ordinary/centered consensus and five deployment-noise repeats. The new quality selector uses 4096 Gaussian repeats and centered consensus; it accesses public arrays alone. Both conditions supply private targets through actual client reports. Private clean counts and benchmark labels serve offline diagnostics, never private configuration selection.

Matched baselines. DP-FedCT uses one hard class vote per proxy example, the same distributed noise layer and the same trainer and consensus correction as SDD. DP-Head clips $128 \times C$ updates relative to the previous broadcast, enforces the integer ball in Lemma 3.2, privately averages, and resets client momentum after each broadcast. Six public-teacher norm quantiles select its clipping radius before private evaluation. Its reported global and local endpoints are distinct. PATE-DI receives protected confidence screening and capped answers, with both costs charged through data-independent Rényi accounting; Appendix D specifies its richer secure functionality. All primary methods protect replace-one whole clients at $\delta = 10^{-5}$, fixed participation and actual/calibrated n.

Replications and paired estimates. Original full grids use seeds 801–803; five-seed controls use 801–805. New decoder confirmation uses 8201–8205, held separate from the original five; all four decoders were frozen before these new outcomes. The 1k decoder check uses the same five fresh seeds. Natural writers use 8401–8405. The Web decoder check replays five existing fixed-$K = 6$

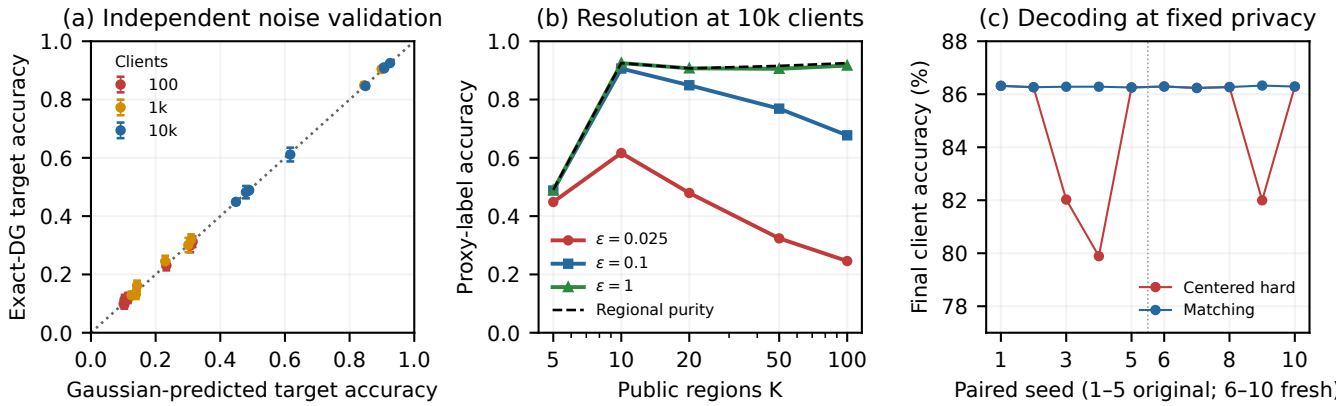


Figure 2: Release-quality and decoding mechanisms. (a) 27 independent exact-noise conditions, 128 repeats each; bars are 1.96 **Monte Carlo standard errors. Predictions use 4096 independent Gaussian repeats and centered consensus. (b) Conditional first-release proxy-label quality at 10k clients, seed 8301; these curves concern released targets. (c) Final CIFAR-10 client accuracy, fixed** $K = 10$, $\varepsilon = 0.1$**, original/fresh five-seed groups. Matching changes decoding of the same protected reports.**

seeds, 8101–8105; the original public-configured confirmation retains all ten seeds 8101–8110. For paired seed differences d_j, intervals are $\bar{d} \pm t_{0.975,m-1} s_d/\sqrt{m}$; tables show mean and sample SD. No private outcome routes a client or seed to a decoder. Full schedules, per-region matrices, all variants and original results remain in the artifact.

5.2 Predicting regional release quality

The quality model accurately describes the object it predicts. A frozen CIFAR-10 diagnostic grid spans actual $n = 100, 1000, 10000$, five values of K, seven privacy budgets and three independent seeds, giving 315 first-release conditions. Each uses the real independently trained cohort's clean vote matrix. These private matrices are offline diagnostic inputs, separate from the public deployment rule.

At 27 prespecified conditions (three cohorts, $K = 5, 10, 20$ and $\varepsilon = 0.025, 0.1, 1$), 128 independent exact distributed-noise releases check 4096 Gaussian simulations. Centered target accuracy has mean absolute error **0.46 percentage points**; 26/27 conditions lie within two combined Monte Carlo standard errors. Uncentered decoding has 0.54-point error. Figure 2(a,b) separates noise-supported target decisions from the regional purity envelope. It concerns the first release, before adaptive training can change the vote margins.

The public-only quality selector chooses $K = 5$ for the smallest cohorts and budgets, moving to $K = 10$ as the noise scale permits. Across the 27 checked cohort/budget/seed combinations, its selected K also maximizes conditional predicted proxy-label quality among the five candidates under the private diagnostic matrices. This validates a public design surrogate for this grid; final local-model accuracy is measured independently below.

5.3 Client-sourced supervision at taxonomy resolution

With $K = C = 10$ and no public task labels, weighted matching reaches 86.3% **CIFAR-10 accuracy using 100 report coordinates** at $\varepsilon = 0.1$. The original five paired seeds give 86.28% ± 0.02%; five independent seeds give 86.28% ± 0.03% (Table 2). On the original five, DP-FedCT gives 69.7% with 20000 coordinates and independent local learning 42.4%. The paired gains are 16.6 points ([14.6, 18.5]) over per-example transfer and 43.9 ([43.8, 44.0]) over local learning. One regional target supervises every proxy item in its region, yielding the 200× report reduction at fixed whole-client budget and local computation.

Class allocation explains the seed differences. Centered hard decoding gives 84.2% ± 3.0% on the original seeds and 85.4% ± 1.9% on fresh seeds. Two original and one fresh seed release nine distinct classes over ten regions in round 1. The repeated-class targets persist into later rounds; their first-release proxy-label accuracies are 83.2/84.1%, versus regional purity 92.5%. Matching maps these three seeds to the full-coverage regional assignment, whose proxy-label accuracy is 92.5%, and leaves the other seven final endpoints unchanged (Figure 2(c)). The ten-seed paired mean difference is 1.5 points with interval $[-0.3, 3.3]$; the mechanism evidence is the exact repair of the three affected trajectories, alongside the separately reported seed uncertainty. Projected and centered soft targets give the other prespecified allocations in Table 2. The released histograms represent hard-vote frequencies; their soft values need not provide the same training signal as a coverage-constrained hard assignment.

5.4 Semantic geometry and decoder conditions

Semantic grouping supplies more than a smaller query count. In the original public-configured five-seed CIFAR-10 control, identical report sizes, budgets, learner features and training give semantic means 85.0% versus 48.9% for size-matched random regions: a

Table 2: CIFAR-10, actual 10k clients, $\varepsilon = 0.1$, fixed $K = C = 10$, no public task labels. Accuracy (%, mean±sample SD); complete means all ten classes occur among first-release regional targets. Every prespecified decoder and both disjoint seed groups are retained.

Decoder	Original five	Fresh five	Complete orig.	Complete fresh
Centered hard	84.15±3.01	85.42±1.91	3/5	4/5
Weighted matching	86.28±0.02	86.28±0.03	5/5	5/5
Projected soft	74.02±1.47	74.52±2.64	4/5	5/5
Centered soft	79.82±1.63	79.16±4.21	3/5	4/5

36.1-point paired gain ([32.0, 40.1]). Public purities are 0.925 and 0.139. Changing the partition alone to pixel-PCA gives 54.9% at purity 0.240; using semantic centroid representatives gives 83.3% (Table 5).

A fresh fixed-rule control uniformly selects ten public queries, collects one vote per query and propagates each target to its public Voronoi cell. It uses the same 100-coordinate interface and noise budget, giving 58.9%±2.3% over the five new seeds. Semantic mean regions with matching gain 27.4 points ([24.6, 30.2]); even centered regional decoding gains 26.5 ([22.7, 30.4]). Public semantic partitioning therefore improves target placement at fixed query count and propagation resources.

Coverage is a conditional decoding choice. At $K = C$, class coverage is a combinatorial guarantee; geometric class alignment is an empirical condition. Fresh MNIST seeds give centered hard 53.3% and matching 52.5%. In the fixed six-region Web check, centered hard gives 19.5% balanced accuracy and matching 14.9%. These tasks favor the unconstrained allocation under the tested geometry; Table 6 retains every decoder. At actual 1k CIFAR-10 clients, centered/matching means are 50.6/52.2% with sample SD 11.5/10.2 points across the fresh five. Matching enforces coverage without supplying the regional vote evidence that a smaller cohort must provide. These controls motivate specifying a decoder with its geometric assumption, rather than routing from private test accuracy.

5.5 Natural clients and supervision allocation

Natural clients test the same interface without duplicating an author or writer to enlarge n. In the ten-seed public-configured Usenet study, $K = 5$ regional representatives reach 26.3% balanced accuracy with 3000 distinct authors at $\varepsilon = 1$. Per-example transfer gives 21.3%: the paired gain is 5.1 points ([3.6, 6.5]), using 30 rather than 9000 coordinates. Semantic assignment adds 8.1 points ([7.0, 9.2]) over matched random representatives. Mean regional votes give 23.9% under the same K, budget and partition. All ten frozen seed outcomes and full class-recall profiles are retained (the artifact's per-family tables).

With five regions for six families, each hard regional release contains at most $\min(K, C) = 5$ distinct class targets. In releases 1/2, representatives allocate 98.7% of proxy supervision to comp/rec/sci, while mean opinions allocate 46.1% to talk/alt–soc/misc. Per-run minimum recalls are 1.8/2.3% for R/M. Both estimators' macro gains coexist with these explicit allocations; private local examples and changing round targets also supervise final models. The fixed $K = C = 6$ ten-seed condition gives M 20.3%, R 17.6%, DP-FedCT 21.3% and Local-only 17.0%. Its different resolution and selection resources are reported separately.

The 62-class natural-writer experiment identifies a different supported resolution. Its fixed $K = 62$ public partition has purity 0.430, compared with CIFAR-10's 0.925 at $K = 10$. This is a directly measured property of the supplied public geometry; the feature extractor and downstream training resources are held to the same specification. Table 3 retains both budgets and all release interfaces at 2000 writers; Table 7 adds 1000 writers. Each client contains its own original records, and evaluation writers are disjoint.

At 2000 writers and $\varepsilon = 1$, matching gives 33.5% accuracy versus 33.4% for per-example transfer, a paired difference of 0.1 points with interval $[-0.3, 0.5]$. It uses 3844 rather than 124000 coordinates, a 32.3× smaller report. Its macro recall is 12.0% versus 11.1%, a paired gain of 0.88 points ([0.73, 1.02]). This operating point connects compact regional supervision to a natural, nonoverlapping image population. At $\varepsilon = 0.1$, the same table records the allocations of each decoder alongside independent learning and the two head endpoints. Both accuracy and macro recall are retained because the 62-class distribution is uneven.

These natural studies distinguish two properties of coverage. Matching ensures that every class is represented among regional targets; it does not assert equal recall of final local models or a one-to-one semantic taxonomy in the public regions. The writer data and fixed Web decoder controls test that distinction directly. Public geometry determines what can be shared as a constant regional target, while private examples and the released target sequence jointly determine the final client's predictions.

5.6 Choosing the released object with available resources

Release resolution is one part of the choice; parameter averaging is another use of the same public representation. At $10k/\varepsilon = 0.1$, the original public-configured regional interface gives 85.0% with 100 coordinates, while DP-Head gives 91.5% at its matched local endpoint and 92.7% globally with 1280 coordinates. These are distinct accuracy–report-size operating points, with the head providing higher accuracy when that release is supported by the cohort and budget.

At $1k/\varepsilon = 0.1$, the independent ten-seed public-configured estimate is 45.6% with 200 regional coordinates, versus 27.2% local and 23.1% global for DP-Head. The paired local-head gain is 18.4 points ([12.1, 24.8]); DP-FedCT gives 46.7% with 20000 coordinates. The original full budget curves and actual-cohort comparisons retain these crossovers (Figures 1 and 3). The regional target model

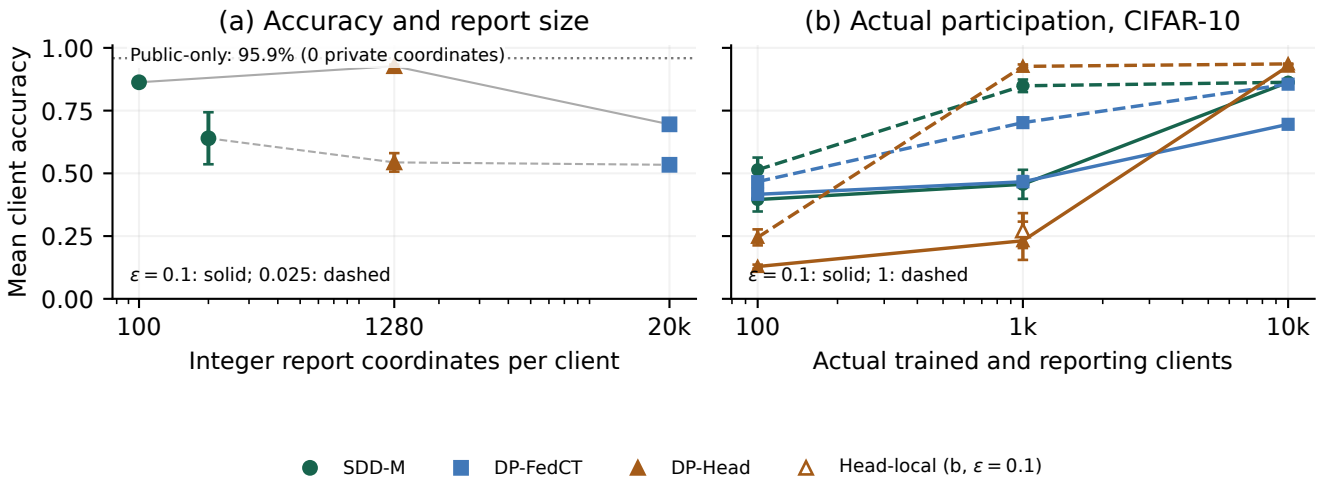


Figure 3: Original public-configured release-resource operating points, CIFAR-10 $\alpha = 0.1$, fixed client steps. (a) Actual 10k clients, three seeds; solid/dashed budgets $0.1/0.025$. (b) Actual cohorts; solid/dashed budgets $0.1/1$. The independent ten-seed confirmation supplies 1k/0.1 points. Open triangle: DP-Head local endpoint. Dotted line: public-only head with 5000 labels, 600 steps and zero private coordinates. Report counts precede cryptographic overhead.

Table 3: Federated EMNIST: 2000 distinct private writers, 62 classes, five seeds and 180 steps. Each budget shows accuracy / macro recall (%, mean±sample SD). No writer or source example is duplicated between clients; held-out writers are disjoint. SDD fixes $K = C = 62$; DP-Head clipping uses public development labels.

Method	Coords.	0.1: acc.	0.1: macro	1: acc.	1: macro
Local-only	0	32.15±0.03	11.95±0.01	32.15±0.03	11.95±0.01
SDD	3844	32.19±0.31	11.32±0.18	32.56±0.49	10.36±0.15
SDD-Match	3844	31.78±0.37	11.45±0.08	33.53±0.24	12.01±0.09
DP-FedCT	124000	31.74±0.11	11.69±0.02	33.43±0.15	11.14±0.06
DP-Head global	7936	10.92±4.90	4.18±1.58	31.18±0.85	9.52±0.28
DP-Head local	7936	13.90±4.65	4.99±1.30	31.42±0.85	9.63±0.30

diagnoses geometry and noise; it does not claim a universal analytic boundary against parameter optimization.

Public labels also support direct training. The same labelled development pool available to the public pilot gives a public-only CIFAR-10 head 95.6/95.9% at 180/600 steps; MNIST gives 84.9/89.0%, and Web gives 65.6/68.2% balanced accuracy. These references use zero private report coordinates. Public-source proxy labels give local CIFAR-10 87.0% and Web 41.0% (Table 4). The fixed experiments measure client-sourced supervision when that labelled-public training resource is absent. Public feature quality, labelled-public resources, cohort size and report granularity are therefore explicit parts of the release choice.

6 CONCLUSION

SDD makes semantic regions a controllable unit of client-private supervision. The KC-coordinate report reuses one target across related examples, with tight sensitivity $\sqrt{2K}$. A regional-label quality model connects geometry, vote margins and privacy noise; paired decoding experiments identify the role of class allocation. With strong frozen public features, 10k CIFAR-10 clients and $\varepsilon = 0.1$, taxonomy-level matching reaches 86.3% without public task labels and uses 200× fewer coordinates than per-example transfer. Natural clients and compact-head comparisons make the available supervision resources explicit. Release granularity is a measurable design choice for learning through protected aggregates.

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A SUPPLEMENTARY RESULTS

All original numerical conditions remain in the artifact, including complete budget grids, seed reconciliation, original-architecture checks and the unmodified original figures. The tables below retain supervision-source references, geometry controls, all new decoder variants and natural-writer accuracy/macro recall. Every new per-region count and released target is indexed as an offline diagnostic, outside the deployment transcript.

B PROOFS AND INTEGER CALIBRATION

For h independent integer-Gaussian shares, the exact correction used in Theorem 3.3 is

$$\tau_h(b) = 10 \sum_{j=1}^{h-1} \exp\left(-2\pi^2 b^2 \frac{j}{j+1}\right). \tag{11}$$

Table 4: Public-configured supervision-source and parameter-endpoint references, percent mean ± sample SD (Web: balanced accuracy). Images use $\alpha = 0.1$, $\varepsilon = 0.1$, five seeds at 10k and ten elsewhere; Web uses ten. M/R denote image/Web SDD. All share public features. Pub heads use all 5000 image/1500 Web development labels, learning rate 0.3 and zero private report coordinates; Local+Pub uses their proxy targets. Client methods use 180 steps. Head-local precedes the third aggregation, retaining two broadcasts with all three releases charged.

	supervision sources					parameter endpoints	
task / variant, cohort	SDD	Local	Local+Pub	Pub (180)	Pub (600)	Head-global	Head-local
CIFAR (M), 10k	85.0 ± 2.9	42.4 ± 0.1	87.0 ± 0.0	95.6 ± 0.1	95.9 ± 0.1	92.7 ± 0.5	91.5 ± 0.4
MNIST (M), 10k	55.0 ± 1.1	33.3 ± 0.1	64.1 ± 0.1	84.9 ± 0.1	89.0 ± 0.1	67.3 ± 1.0	65.0 ± 1.0
CIFAR (M), 1k	45.6 ± 5.8	42.3 ± 0.4	87.0 ± 0.1	95.6 ± 0.1	95.9 ± 0.1	23.1 ± 7.6	27.2 ± 7.0
Usenet (R), 1k, 0.25	24.8 ± 3.0	17.1 ± 0.0	41.0 ± 0.4	65.6 ± 0.4	68.2 ± 0.5	22.7 ± 4.7	22.8 ± 2.6
Usenet (R), 3k, 1	26.3 ± 1.6	17.0 ± 0.0	41.0 ± 0.3	65.6 ± 0.4	68.2 ± 0.5	16.8 ± 0.1	18.3 ± 0.3

Table 5: Semantic attribution at actual $n = 10^4$, $\alpha = 0.1$, $\varepsilon = 0.1$, five seeds. Purity is public; pseudo-label accuracy and flips use the first release. Full controls at both budgets and heterogeneity levels are in the artifact.

task	grouping	K	purity	pseudo	flips	accuracy
CIFAR	SDD	10	0.925	0.906	0.020	0.850 ± 0.029
CIFAR	random	10	0.139	0.121	0.660	0.489 ± 0.009
CIFAR	representative	10	0.925	0.876	0.120	0.833 ± 0.029
CIFAR	weak	10	0.240	0.222	0.380	0.549 ± 0.005
MNIST	SDD	20	0.624	0.552	0.210	0.550 ± 0.011
MNIST	random	20	0.145	0.116	0.680	0.378 ± 0.014
MNIST	representative	20	0.624	0.556	0.200	0.555 ± 0.013
MNIST	weak	20	0.683	0.513	0.340	0.512 ± 0.014

Table 6: Fixed taxonomy resolution without public task labels, five paired seeds. MNIST/CIFAR-10: accuracy at $\varepsilon = 0.1$; Web: balanced accuracy at $\varepsilon = 1$, $K = 6$. – denotes an unrun prespecified cell.

Task/cohort	Centered hard	Weighted matching	Projected soft	Centered soft
mnist, 10k	53.34±1.12	52.53±0.55	47.97±1.38	49.38±2.54
cifar10, 1k	50.61±11.47	52.23±10.24	–	–
web, 3k	19.53±3.61	14.90±1.10	19.19±0.08	18.94±0.59

The empty sum is zero. With $L = \log(1/\delta)$ and $\rho_\varepsilon = (\sqrt{L+\varepsilon} - \sqrt{L})^2$, calibration uses

$$b = \left\lfloor \frac{\Delta_q}{\sqrt{2h\rho_\varepsilon/T}} \right\rfloor + 1. \tag{12}$$

It verifies the full corrected bound after upward rounding.

Proof of Lemma 3.1. Squared Euclidean distance decomposes over rows: $\|R - R'\|_2^2 = \sum_{k=1}^{K} \|R[k,\cdot] - R'[k,\cdot]\|_2^2$. For simplex vectors p, p', $\|p\|_2^2 \le 1$, $\|p'\|_2^2 \le 1$ and $\langle p, p' \rangle \ge 0$, so $\|p - p'\|_2^2 \le 2$. Distinct one-hot rows attain equality. Repeating them across all K rows attains $2K$. Statistical independence between rows is unnecessary. □

Proof of Lemma 3.2. Write $\mathrm{round}(sR) = sR + e$, with $|e_j| \le 1/2$. Then $\|e - e'\|_2 \le \sqrt{d}$, giving $\Delta_q \le s\sqrt{2K} + \sqrt{d}$. One-hot votes have zero rounding error. Two updates clipped to a radius-B ball differ by at most $2B$, giving $2sB + \sqrt{d}$ after rounding. The implemented guard confines integer reports to radius $L_B = \lceil sB + \sqrt{d}/2 \rceil$, hence sensitivity $2L_B$. It checks squared integer norms exactly in the recorded numeric range and replaces an out-of-ball report with zero. Each conditional update is clipped relative to the same preceding public broadcast. □

Proof of Theorems 3.3 and 3.4. Condition on a public transcript and a coalition's disclosed contributions. The remaining integer query has sensitivity Δ_q and retains at least h independent scale-b noise shares per coordinate. Proposition 10 of Kairouz et al. [17] gives the zCDP upper bound in Eq. (4). Its first branch suffices for all reported certificates. Since $j/(j+1) \ge 1/2$, $\tau_h \le 10(h-1)e^{-\pi^2 b^2}$. Modulo reduction and later target construction are post-processing. The same conditional bound holds at every round, so adaptive composition sums the bounds. Equation (5) converts the sum to client-level (ε, δ)-DP. □

Strict numerical slack. The implementation increases the integer scale beyond the Gaussian reference and verifies $\Delta_q^2/(2hb^2) + d\tau_h/4 \le \rho_\varepsilon/T$. It stores $\log[10(h-1)e^{-\pi^2 b^2}]$ and conservatively uses 10^{-30} as an upper bound when the exponential is smaller.

Thus underflow never silently discards the correction. The experiment artifacts certify each stage and its composed budget. For 10% collusion tolerance, both h and its correction are recomputed. Ninety-digit recomputation verifies all composed certificate bounds; the largest retained convolution upper bound is 10^{-30}.

Ring arithmetic. Reports and shares are added modulo 2^{64}; the signed representative is dequantised for learning. Privacy is preserved under this modular map even when wraparound occurs. Utility wraparound is controlled using the sub-Gaussian tail of the discrete Gaussian [17]: for a scalar noise sum G, $\Pr(|G| \geq a) \leq 2e^{-a^2/(2nb^2)}$. A union bound across released coordinates and rounds, with signal bounded by the quantised report range, certifies the recorded configurations. Its log probability is below -3.5×10^{18} across primary releases.

C REPRODUCIBILITY AND PUBLIC SELECTION

The anonymous supplementary archive has a root `README.md` and `reproduce.py`. Run `python reproduce.py verify` to check inputs, frozen configurations, all numerical tables and stored outputs; `python reproduce.py smoke` replays representative fixed cases. `EXPERIMENT_INDEX.csv` maps each condition to its seeds, configuration, role and result. Feature bundles, candidate scores, raw logs, exact integer certificates and complete grid tables are included. Public selection omits test arrays; configuration hashes are checked against the frozen manifests.

The frozen image extractor is DINOv2 ViT-S/14 from the official public checkpoint [24]; its hash is recorded. All images use 224×224 RGB bicubic resizing and the checkpoint's normalisation. PCA and feature normalisation are fixed public preprocessing; application to private examples is point-wise. Public K-means uses five initialisations, at most 100 iterations, seed 1729, and deterministic nonempty-cluster repair. Random grouping preserves the semantic partition's exact region sizes. The feature control changes the partition to raw-pixel PCA while retaining the same client learner and feature map. The representative control queries the proxy point closest to each public centroid.

Public pilot teachers use seed 1729, disjoint development data, and the same round-1 learner as the final experiment. Candidate K and correction modes use mean-posterior opinions and are scored by nearest-public-centroid label transfer on validation data, averaged over noise seeds 91001–91005. Exact ties prefer smaller K, then ordinary consensus. Six DP-Head clipping candidates are public norm quantiles 0.5, 0.75, 0.9, 0.95, 0.99, 1.0; they use five noise repeats. PATE-DI settings use two fixed noise repeats per public candidate. These public design surrogates neither use private client data nor consume their privacy budget. Per-budget choices are retained alongside every candidate score, enabling independent verification of the selection rule. The representative comparisons retain those same K/consensus choices; no private result selects between M and R. The no-public-label check instead uses fixed taxonomy resolution and centered consensus. Proxy labels enter its offline diagnostics, while training targets come from the protected client sum.

New decoder and quality studies. The new frozen schedules separately record original seeds 801–805, fresh confirmation seeds 8201–8205, quality-diagnostic seeds 8301–8303 and natural-writer seeds 8401–8405. Weighted matching maximizes public-size-weighted projected votes at $K = C$. Soft targets use projected vote proportions; centered soft targets use temperature 0.1, fixed before the new outcomes. They use identical private/proxy draws and steps; no test-driven decoder routing is performed. Uniform-query controls select K proxy examples without replacement with seed $1729 + K$, query them once each, and propagate to nearest query cells. This keeps the regional sensitivity and coordinate count.

The new public selector explicitly loads proxy, development and validation arrays; it omits private and test arrays. Gaussian target-quality predictions use 4096 repeats. Exact-noise validation uses a disjoint seed stream and 128 repeats at each of 27 prespecified conditions. Conditional clean private votes are retained for diagnostic validation alone. The first-release grid spans 315 cells; its held-out nature concerns noise replications, with no claim that these conditional predictions estimate final model accuracy. Natural-writer clipping uses the same six public-teacher norm quantiles, scored with 128 Gaussian deployment-noise repeats on public validation. The fixed regional writer runs use $K = 62$ and neither public task labels nor private scores to choose their configuration.

Exact noise and experimental replay. The sampler uses a discrete-Laplace rejection proposal and rational Bernoulli $\exp(-\gamma)$ decisions as in Canonne et al. [9]. Rejection from ideal independent random bits and arbitrary-precision fallbacks retain the exact integer distribution and its tail. The executable uses ChaCha20 pseudorandom bit streams under the corresponding cryptographic random-source assumption, with fresh secret keys drawn from the operating system for deployment. The privacy proof uses independent private random coins. Recorded benchmark seeds reproduce offline utility calculations and diagnostics, rather than defining a publicly seeded deployment privacy transcript. Sampler checks compare zero-event probabilities, moments and tails with analytic integer-Gaussian probabilities; the learner is checked against scalar SGD and one-client noninterference before final tests.

Resource and protocol scope. Experiments use one RTX 5090 GPU and 16 CPU noise-sampling threads. Vectorisation trains every client independently and samples every client's share; cached round-1 states accelerate paired comparisons. Private models are not transmitted. The utility simulator evaluates the noisy sum directly and assumes the protected transport of A1–A4. Reported clean pseudo-label metrics and test accuracies are offline benchmark measurements, outside the claimed deployment transcript.

D BASELINE ACCOUNTING AND TRAINING

For DP-FedCT, the public singleton partition is exactly the per-example vote interface. SDD and this variant share initial local states, private batches, local optimiser, budget conversion and post-processing. For DP-Head, each client clips its update relative to the preceding public global head and enforces the public integer-ball

Table 7: Federated EMNIST, 1000 distinct private writers. Five paired seeds, 180 steps per client; accuracy and macro recall (%, mean±sample SD). Settings match Table 3. Both cohorts and all per-seed outcomes are retained in the artifact.

Writers	ε	Method	Coords.	Accuracy	Macro recall
1k	0.1	Local-only	0	32.21±0.13	11.95±0.04
1k	0.1	SDD	3844	31.48±0.37	11.47±0.04
1k	0.1	SDD-Match	3844	31.61±0.33	11.56±0.10
1k	0.1	DP-FedCT	124000	31.71±0.18	11.68±0.06
1k	0.1	DP-Head global	7936	3.35±0.92	2.33±0.44
1k	0.1	DP-Head local	7936	4.82±1.02	2.86±0.76
1k	1.0	Local-only	0	32.21±0.13	11.95±0.04
1k	1.0	SDD	3844	32.46±0.65	10.34±0.21
1k	1.0	SDD-Match	3844	33.44±0.34	11.85±0.11
1k	1.0	DP-FedCT	124000	32.87±0.16	11.69±0.06
1k	1.0	DP-Head global	7936	30.00±2.85	9.24±0.97
1k	1.0	DP-Head local	7936	30.25±2.31	9.31±0.77

radius after rounding; the server averages the protected integer updates, broadcasts the new head and generates proxy targets from this public model. All clients adopt that head and reset momentum. Each round still has 60 local steps. We evaluate both the final broadcast and the clients immediately before that broadcast, retaining the first two protected heads, with no extra local steps. The third release remains charged for both endpoints.

For PATE-DI, let $v(x)$ be the aggregate class-count vector. Protected screening forms noisy $\max_c v_c(x)$ for all $|P|$ queries. Its replace-one L_2 sensitivity is $\sqrt{|P|}$. Thresholds are $0.1, 0.2, 0.3$ in mean-vote units; answer caps are $0.25|P|$ or $0.5|P|$; screening receives 0.25 or 0.5 of the per-round privacy allowance. The remaining allowance answers at most M queries with joint sensitivity $\sqrt{2M}$. Thresholding and selecting the highest noisy confidences are post-processing of screening. The answer sensitivity holds conditionally on every selected subset. Both stages use the corrected distributed integer-noise bound and compose over T rounds. The tighter data-independent Rényi conversion [4] is

$$\varepsilon \le \inf_{a>1} \left[a\rho_{\text{total}} + \frac{\log(1/(a\delta))}{a-1} + \log(1-1/a) \right]. \tag{13}$$

This conversion is granted to PATE-DI in addition to protected screening. The evaluated view includes noisy confidences and selected noisy count vectors. Data-dependent PATE accounting requires additional protected analysis of private consensus and accounting outputs; it is a distinct protocol scope.

Original architecture. The artifact retains the seed-42 ResNet-18 diagnostic and exact unrounded logs. DP-FedCT's two 25.5% entries round distinct values, 25.4823/25.4830%, at $\varepsilon = 0.1/1$. Their noisy targets and trajectories differ. These actual-$n = 100$ architecture checks are separate from the frozen-feature study.